



Started on	Thursday, 6 August 2026, 10:07 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:29 AM
Time taken	22 mins 9 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main()
3  {
4      int n;
5      int i=1;
6      int s=1;
7      int count=2;
8      scanf("%d",&n);
9      while(s<=n)
10     {
11         count++;
12         i++;
13         count++;
14         s+=i;
15         count++;
16     }
17     count++;
18     printf("%d",count);
19     return 0;
20 }
21
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 6 August 2026, 10:30 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:37 AM
Time taken	6 mins 57 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main()
3  {
4      int n;
5      scanf("%d",&n);
6      int c=0;
7      if(n==1)
8      {
9          c++;
10         // printf("*");
11         c++;
12     }
13     else
14     {
15         c++;
16         for(int i=1;i<=n;i++)
17         {
18             c++;
19             for(int j=1;j<=n;j++)
20             {
21                 c++;
22                 //printf("*");
23                 c++;
24                 //printf("*");
25                 c++;
26                 break;
27             }
28             c++;
29         }
30         c++;
31     }
32     printf("%d",c);
33 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 6 August 2026, 10:37 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:47 AM
Time taken	10 mins 6 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main()
3  {
4      int num;
5      scanf("%d",&num);
6      int c=0;
7      for (int i = 1; i <= num; ++i)
8      {
9          c++;
10         if (num % i == 0)
11         {
12             c++;
13             // printf("%d ", i);
14         }
15         c++;
16     }
17     printf("%d",c);
18     return 0;
19 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Tuesday, 25 August 2026, 12:47 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 12:47 PM
Time taken	21 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  void function(int n)
4  {
5      int c = 0;
6
7      c++; // count the initial execution
8
9      for (int i = n / 2; i < n; i++)
10     {
11         c++; // i < n condition
12
13         for (int j = 1; j < n; j = 2 * j)
14         {
15             c++; // j < n condition
16
17             for (int k = 1; k < n; k = k * 2)
18             {
19                 c++; // k < n condition
20                 c++; // c++ statement
21             }
22
23             c++; // final k < n check
24         }
25
26         c++; // final j < n check
27     }
28
29     c++; // final i < n check
30
31     printf("%d", c);
32 }
33
34 int main()
35 {
36     int n;
37     scanf("%d", &n);
38
39     function(n);
40
41     return 0;
42 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Wednesday, 16 September 2026, 7:07 PM
State	Finished
Completed on	Wednesday, 16 September 2026, 7:09 PM
Time taken	2 mins 1 sec
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, count = 3;
6
7      scanf("%d", &n);
8
9      while (n != 0)
10     {
11         count += 4;
12         n /= 10;
13     }
14
15     printf("%d", count);
16
17     return 0;
18 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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